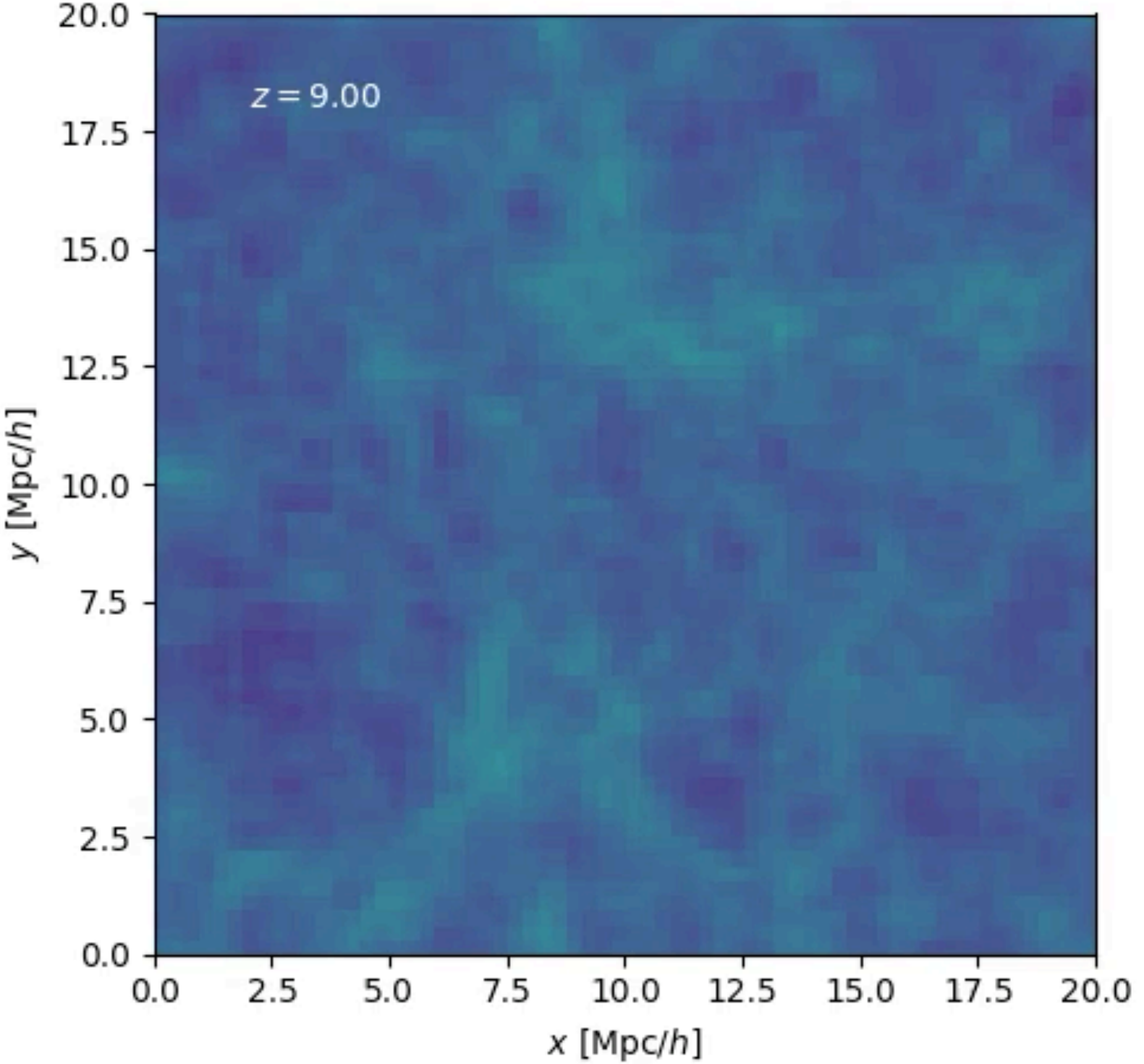
Forward model

8

Observations : Galaxy catalogues → galaxy 3D density maps

Aim : Infer current and past *dark matter* density map from current *galaxy* density map

Method : Using `JAX / Tensorflow`, optimise differentiable N-body simulations (gravitation only ≡ dark matter)

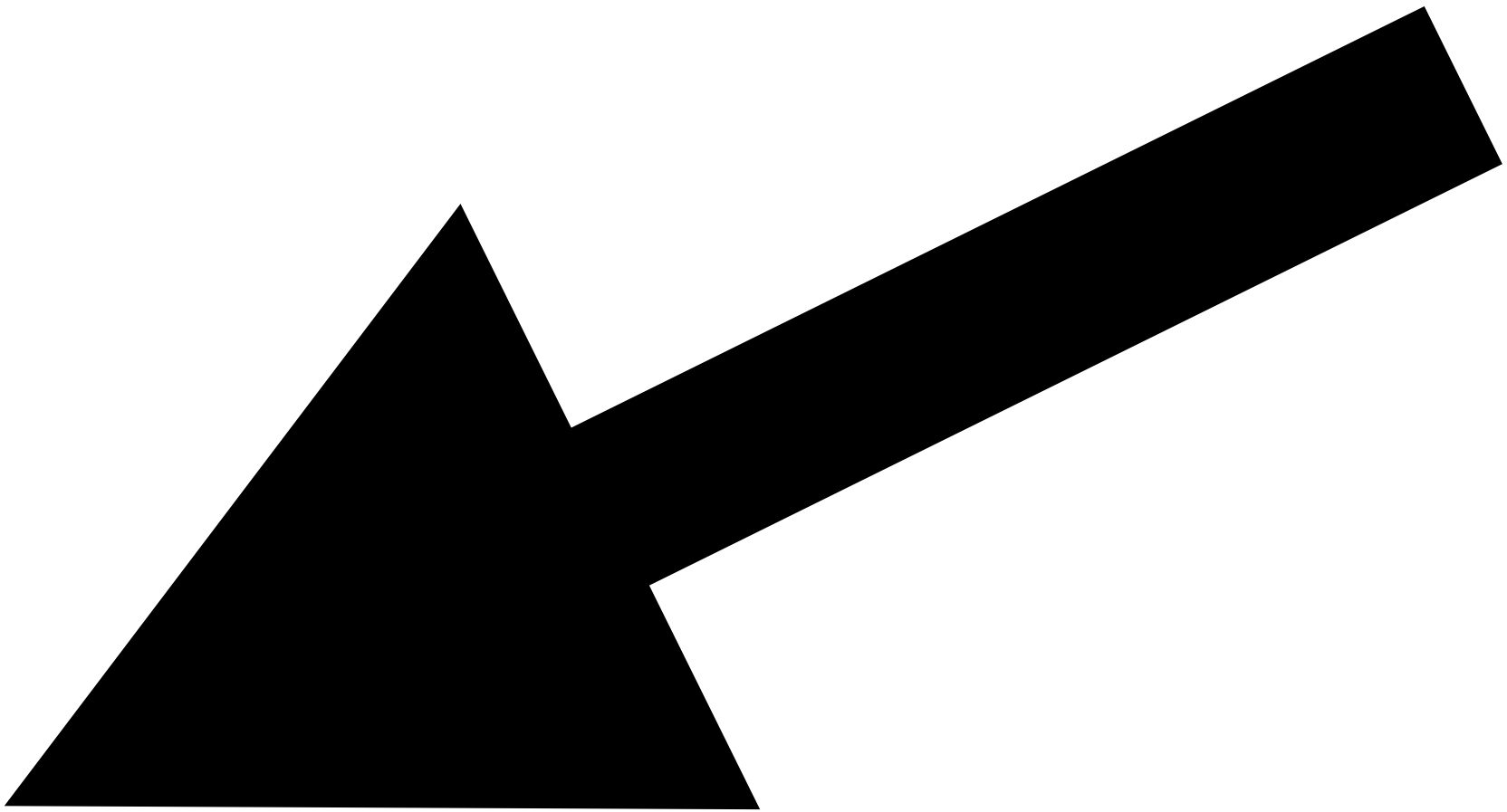

$z = 9.00$
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
y [Mpc/h]
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
x [Mpc/h]

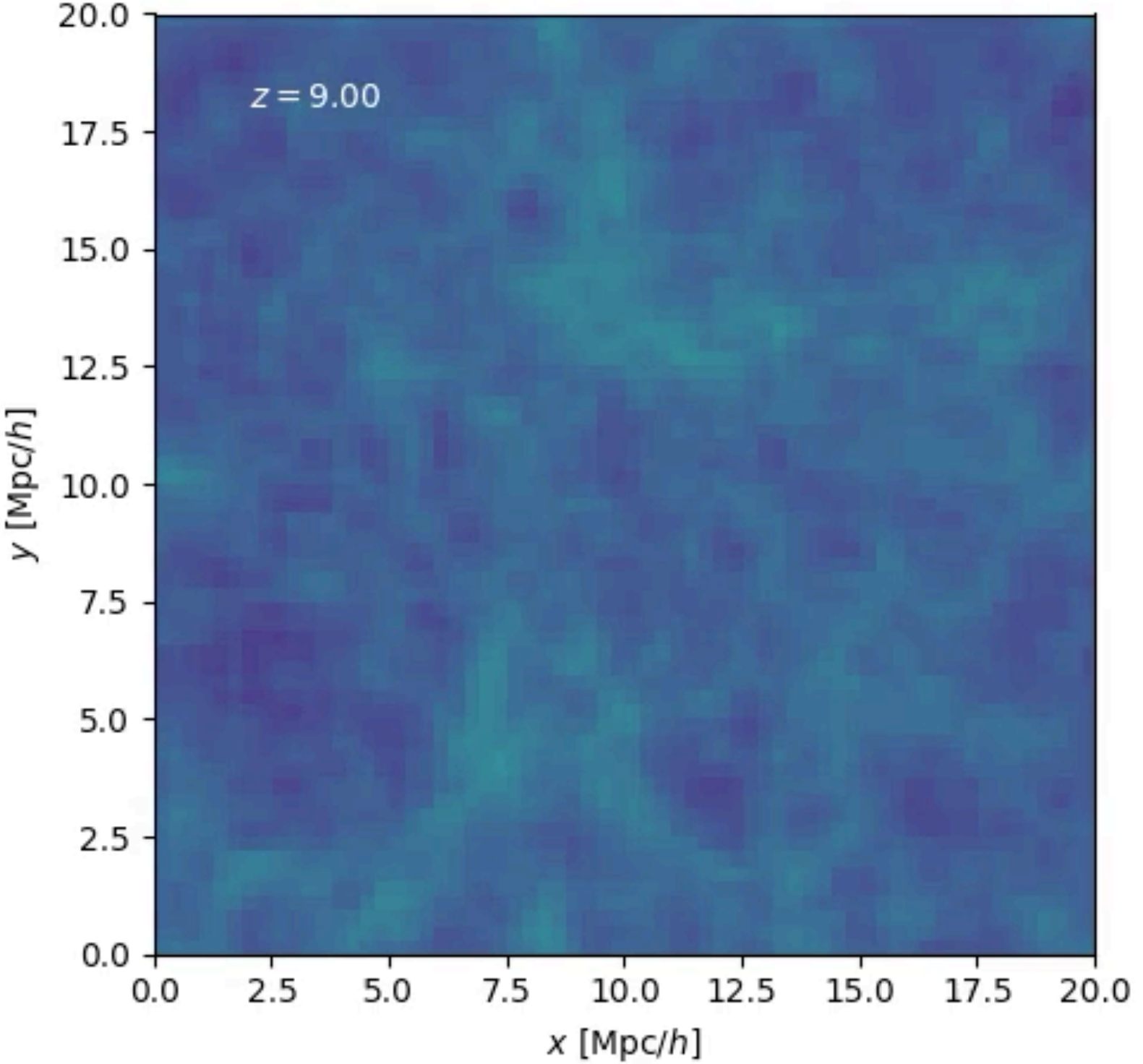
Starting from a realisation of a Gaussian Random Field, we run a N-body simulation that simulates the gravitational collapse of matter in the Universe through cosmic time.

Here on the left is represented a projection of the dark matter density. We can see cosmic structures forming through cosmic time.

The black arrow shows a massive galaxy cluster.

Thanks to GPU acceleration, such a simulation runs in ~0.2 s and is differentiable (10x speed-up, Modi+20)



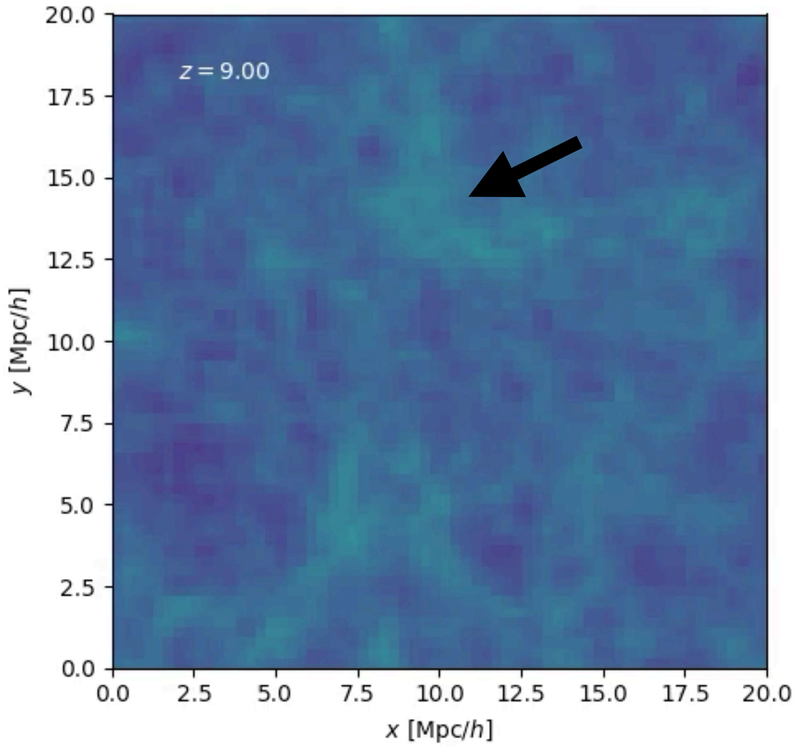

$z = 9.00$
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
y [Mpc/h]
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
x [Mpc/h]

Forward model

Observations : Galaxy catalogues → galaxy 3D density maps

Aim : Infer current and past *dark matter* density map from current *galaxy* density map

Method : Using `JAX / Tensorflow`, optimise differentiable N-body simulations (gravitation only ≡ dark matter)



Starting from a realisation of a Gaussian Random Field, we run a N-body simulation that simulates the gravitational collapse of matter in the Universe through cosmic time.

Here on the left is represented a projection of the dark matter density. We can see cosmic structures forming through cosmic time.

The black arrow shows a massive galaxy cluster.

Thanks to GPU acceleration, such a simulation runs in ~0.2 s and is differentiable (10x speed-up, Modi+20)

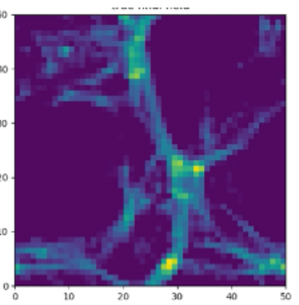
Optimisation

Observations : Galaxy catalogues → galaxy 3D density maps

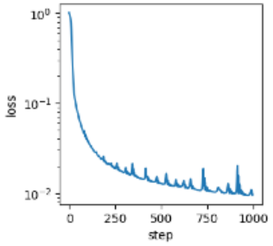
Aim : Infer current and past *dark matter* density map from current *galaxy* density map

Method : Using `JAX / Tensorflow`, optimise differentiable N-body simulations (gravitation only ≡ dark matter)

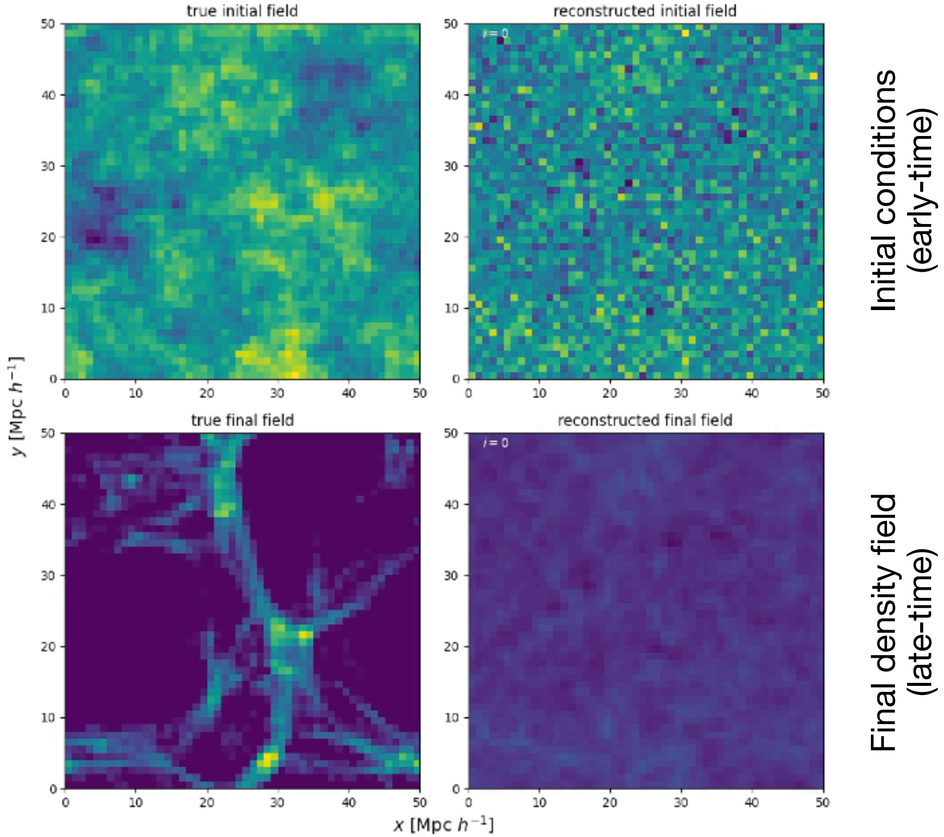
Observation



Loss



Optimisation (ADAM)



The combined speed and differentiability allows to run optimisation and reconstruct the structure formation history from late-time observations (local Universe)

Here I show a simple test implementation of such an optimisation.

We see the constraints on the density field getting better through the optimisation